

ZAMYN-UUD, Mongolia

WMO: 443580

Lat: 43.703N

Lon: 111.901E

Elev: 3163

StdP: 13.09

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.5	-15.6	-25.8	1.5	-14.9	-22.1	1.8	-12.4	26.8	18.5	22.8	11.2	2.9	90	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	94.1	66.6	90.9	65.4	87.8	64.1	71.7	87.5	69.4	84.7	67.5	81.6	8.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	111.0	79.0	64.6	102.4	76.7	62.6	95.6	75.5	37.8	88.1	35.6	85.2	34.0	81.7	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.0	22.2	19.9	-23.4	98.9	4.3	3.3	-26.5	101.3	-29.1	103.3	-31.5	105.1	-34.7	107.5		
			-23.7	73.2	4.4	4.2	-26.8	76.3	-29.4	78.7	-31.9	81.1	-35.1	84.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	1.2	11.1	28.3	45.9	58.8	70.1	75.8	72.0	60.2	43.0	22.4	6.1	(6)
(7)		DBStd	27.17	9.07	10.60	10.39	9.68	8.51	6.61	5.44	5.83	7.79	9.04	10.91	9.17	(7)
(8)		HDD50	5934	1513	1088	673	189	20	0	0	0	12	251	827	1361	(8)
(9)		HDD65	9430	1978	1508	1137	576	227	27	1	10	180	683	1277	1826	(9)
(10)		CDD50	2796	0	0	1	65	295	602	801	682	317	33	0	0	(10)
(11)		CDD65	817	0	0	0	2	37	179	337	227	35	0	0	0	(11)
(12)		CDH74	9021	0	0	0	58	620	2000	3687	2235	413	8	0	0	(12)
(13)		CDH80	3613	0	0	0	10	201	779	1676	845	102	0	0	0	(13)
(14)	Wind	WSAvg	7.0	5.8	6.4	7.8	8.9	9.3	7.2	6.3	6.0	6.7	6.9	6.7	6.3	(14)
(15)	Precipitation	PrecAvg	4.6	0.0	0.0	0.1	0.2	0.4	0.7	1.2	1.1	0.4	0.2	0.1	0.1	(15)
(16)		PrecMax	10.0	0.6	0.2	0.4	0.9	2.2	2.7	6.6	3.1	1.7	1.4	0.7	0.3	(16)
(17)		PrecMin	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.9	0.1	0.0	0.1	0.2	0.4	0.5	1.1	0.7	0.4	0.3	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.6	45.4	64.6	80.8	90.8	95.9	99.9	95.7	87.7	74.4	55.0	36.5	(19)
(20)			MCWB	25.5	35.6	43.8	55.2	60.1	64.9	68.9	69.1	61.2	55.6	41.9	28.7	(20)
(21)		2%	DB	25.1	38.9	56.7	73.9	85.7	91.8	95.4	91.5	82.9	68.4	48.8	28.7	(21)
(22)			MCWB	20.3	30.3	41.2	51.8	59.1	64.3	67.6	66.9	60.3	52.3	38.5	23.5	(22)
(23)		5%	DB	20.6	33.9	51.4	68.7	80.9	88.0	92.0	88.0	78.6	63.6	43.5	24.1	(23)
(24)			MCWB	16.9	27.0	38.0	49.1	56.4	63.2	66.5	65.2	58.3	48.8	34.8	19.9	(24)
(25)		10%	DB	16.1	29.1	46.0	63.8	76.4	84.4	88.7	84.6	74.8	58.6	38.7	20.3	(25)
(26)			MCWB	13.3	23.7	34.6	46.5	54.7	62.4	65.5	63.8	56.9	45.5	31.7	17.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	36.2	46.0	57.7	65.2	70.9	75.7	74.1	66.7	57.5	43.6	29.2	(27)
(28)			MCD B	31.1	45.3	60.9	76.1	86.0	87.8	92.8	89.8	81.3	71.7	53.7	36.3	(28)
(29)		2%	WB	20.6	30.6	42.0	53.5	61.2	68.6	72.5	71.3	63.4	53.4	38.6	23.9	(29)
(30)			MCD B	25.0	38.1	56.0	70.3	79.1	85.6	89.0	86.2	77.1	66.3	48.3	28.7	(30)
(31)		5%	WB	17.0	27.3	38.3	50.3	58.2	66.5	70.1	68.7	60.9	49.5	35.3	20.4	(31)
(32)			MCD B	20.6	33.8	50.7	66.3	76.2	82.4	85.5	82.9	74.1	61.6	43.3	24.2	(32)
(33)		10%	WB	13.3	23.6	34.9	47.1	56.0	64.6	68.2	66.5	58.4	46.4	32.1	17.2	(33)
(34)			MCD B	16.1	29.0	45.7	62.8	73.4	79.3	82.2	79.4	71.7	57.5	38.8	20.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.3	24.8	25.2	24.7	24.6	22.0	20.7	20.9	22.7	22.8	21.4	19.5	(35)
(36)			MCDBR	25.4	30.6	31.5	31.4	31.7	27.3	25.5	25.1	27.4	29.0	26.0	22.9	(36)
(37)			MCWBR	21.5	23.3	20.7	18.8	17.1	12.6	9.9	10.9	13.5	17.5	18.2	18.8	(37)
(38)		5% WB	MCD BR	25.2	29.7	30.8	29.0	28.4	23.5	21.4	22.0	23.4	27.7	25.1	22.9	(38)
(39)			MCWBR	21.5	22.9	21.1	18.8	17.1	13.2	10.9	11.7	14.1	18.0	18.1	19.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.301	0.342	0.393	0.438	0.466	0.436	0.421	0.382	0.335	0.289	0.271	(40)	
(41)		taud	2.283	2.200	2.126	2.031	1.959	1.982	2.130	2.138	2.175	2.241	2.280	2.255	(41)	
(42)		Ebn at Noon	272	280	282	275	265	258	264	264	265	264	261	258	(42)	
(43)		Edh at Noon	26	35	43	51	57	56	48	46	41	33	26	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	687	1051	1573	1978	2181	2111	2089	1933	1610	1198	813	586	(44)	
(45)		RadStd	82	78	99	95	110	123	85	74	62	59	60	76	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	-1.78	N/A	-0.95	-1.31	N/A	N/A	N/A	N/A

Nomenclature: See separate page